

PAPER_1140: SCm Vacuum Manifold —\Mizuno LENR Transmutation Mechanism

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Abstract

Mizuno LENR experiments (Pd-D and Ni-H gas-loading) report anomalous excess heat and elemental transmutation products without the hard radiation expected from nuclear reactions. We derive both phenomena from the SCm Vacuum Manifold using the $F_{U,Bi,i}$ buoyancy field, 1.25 THz phonon resonance, and the 26D Ramanujan amplification factor $S_{26}^{(3)} = 1.4531 \times 10^{26}$.

Canonical Constants

Constant	Value	Description
E_{phonon}	$8.28 \times 10^{-22} \text{ J}$	THz phonon energy
$S_{26}^{(3)}$	1.4531×10^{26}	26D Ramanujan amplification
Φ_{res}	0.84	Resonance coupling
β_i	0.6	Buoyancy coefficient
κ	$5 \times 10^{-4} \text{ day}^{-1}$	SCm decay rate
ρ_{SCm}	$7.09 \times 10^{-37} \text{ J/m}^3$	SCm vacuum density
ρ_{UA}	$7.09 \times 10^{-36} \text{ J/m}^3$	UA vacuum density

Mizuno Excess Heat (SCm)

Mizuno observed 10–300~W excess heat in gas-loaded Ni-D/H systems with anomalous transmutation products (Cu, Cr, Fe from Ni). The SCm prediction follows the same phonon pathway as Parkhomov but with a lower cluster density N_M :

$$P_{\text{Mizuno}} = N_M \cdot \varepsilon_{\text{cluster}} \cdot e^{-\kappa t} \cdot f_b$$

where $\varepsilon_{\text{cluster}} = 630 \text{ eV} = 1.009 \times 10^{-16} \text{ J}$ is the canonical Holmlid-calibrated cluster energy, and f_b is the system-specific buoyancy stabilisation factor. For $N_M \sim 10^{20}\text{--}10^{21}$:

$$P_{\text{Mizuno}} \approx 10\text{--}300 \text{ W}$$

Transmutation Mechanism

Standard electroweak theory cannot explain transmutation at low temperatures without MeV-scale particle exchange. SCm provides the mechanism: 1. The 1.25 THz phonon ($E_{\text{phonon}} = 8.28 \times 10^{-22} \text{ J}$) drives coherent oscillation of the SCm vacuum manifold within the metal lattice. 2. $F_{U,Bi,i}$ buoyancy modifies the effective nuclear potential barrier via the $\cos(\pi t_n)$ negative-time term, allowing sub-barrier transmutation. 3. The 26D Ramanujan factor $S_{26}^{(3)}$ amplifies the transition probability without requiring high-energy intermediaries. 4. Energy is released into the phonon bath (heat) rather than into particle channels (no hard radiation), consistent with Mizuno’s calorimetry.

Unified LENR Picture

Experiment	System	Observed P	SCm Prediction
Holmlid	K-Fe catalyst	630 eV KER	$\varepsilon_{\text{cluster}} = 630 \text{ eV}$
Parkhomov	Ni-H, 1100°C	150–280 W	$N \sim 2 \times 10^{18}, f_b = 1$
Pons–Fleischmann	Pd–D	1–50 W	$V = 10^{-6} \text{ m}^3, f_b = 0.001$
Mizuno	Ni–D gas	10–300 W	$N \sim 10^{20}\text{--}10^{21}, f_b \text{ scaled}$

All four experiments are unified under a single equation (Eq.):

$$P_{\text{LENR}} = N_{\text{eff}} \cdot \varepsilon_{\text{cluster}} \cdot e^{-\kappa t} \cdot f_b(\text{system})$$

Conclusion

SCm provides a single first-principles mechanism — phonon resonance amplified by the 26D vacuum manifold and stabilised by $F_{U,Bi,i}$ buoyancy — that quantitatively reproduces all four major LENR experimental results without ad hoc parameters beyond those already calibrated ($\kappa, [SSq], \beta_i, \Phi_{\text{res}}$). The unified heat equation with canonical $\varepsilon_{\text{cluster}} = 630 \text{ eV}$ and system-appropriate N_M and f_b spans the full experimental range from Pons–Fleischmann (1–50 W) to Mizuno (10–300 W) and Parkhomov (150–280 W).